

Power BI Assignment 2 – DAX & Data Visualization

DAX Calculations

Calculated Columns

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

The screenshot displays the Power BI Desktop interface. The 'Table tools' ribbon is active, showing the 'Column tools' tab. The formula bar at the top contains the DAX formula: `Category Type = 'Order Details'[Category] & "-" & 'Order Details'[Sub-Category]`. Below the formula bar, a table view of the 'Order Details' table is shown. The table has columns: Amount, Profit, Quantity, Category, Sub-Category, and Category Type. The 'Category Type' column contains concatenated values of 'Category' and 'Sub-Category' separated by a hyphen. The 'Data' pane on the right shows the 'Order Details' table with 'Category Type' highlighted.

Amount	Profit	Quantity	Category	Sub-Category	Category Type
561	212	3	Clothing	Saree	Clothing-Saree
119	-5	8	Clothing	Saree	Clothing-Saree
193	-166	3	Clothing	Saree	Clothing-Saree
157	5	9	Clothing	Saree	Clothing-Saree
75	0	7	Clothing	Saree	Clothing-Saree
25	-5	4	Clothing	Saree	Clothing-Saree
43	0	3	Clothing	Saree	Clothing-Saree
160	-59	2	Clothing	Saree	Clothing-Saree
1603	0	9	Clothing	Saree	Clothing-Saree
353	90	8	Clothing	Saree	Clothing-Saree
534	0	3	Clothing	Saree	Clothing-Saree
149	-87	4	Clothing	Saree	Clothing-Saree
635	-349	5	Clothing	Saree	Clothing-Saree
24	-9	4	Clothing	Saree	Clothing-Saree
711	-8	4	Clothing	Saree	Clothing-Saree
382	30	3	Clothing	Saree	Clothing-Saree
637	113	5	Clothing	Saree	Clothing-Saree
122	-47	4	Clothing	Saree	Clothing-Saree
20	-8	2	Clothing	Saree	Clothing-Saree
42	-6	4	Clothing	Saree	Clothing-Saree
55	-26	4	Clothing	Saree	Clothing-Saree

Category Type = 'Order Details'[Category] & "-" & 'Order Details'[Sub-Category]

Created a calculated column named '**Category Type**' in the Order Details table by combining the '**Category**' and '**Sub-Category**' columns into a single column.

- Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

File Home Help Table tools Column tools Share

Name **Revenue** % Format Whole number Summarization Sum Data type Whole number \$ % 0 0 Data category Uncategorized Sort by column Data groups Manage relationships New column

Structure Formatting Properties Sort Groups Relationships Calculations

1 Revenue = 'Order Details'[Amount] * 'Order Details'[Quantity]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683
B-25602	119	-5	8	Clothing	Saree	Clothing-Saree	952
B-25603	193	-166	3	Clothing	Saree	Clothing-Saree	579
B-25604	157	5	9	Clothing	Saree	Clothing-Saree	1413
B-25605	75	0	7	Clothing	Saree	Clothing-Saree	525
B-25609	25	-5	4	Clothing	Saree	Clothing-Saree	100
B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320
B-25613	1603	0	9	Clothing	Saree	Clothing-Saree	14427
B-25619	353	90	8	Clothing	Saree	Clothing-Saree	2824
B-25622	534	0	3	Clothing	Saree	Clothing-Saree	1602
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596
B-25625	635	-349	5	Clothing	Saree	Clothing-Saree	3175
B-25628	24	-9	4	Clothing	Saree	Clothing-Saree	96
B-25633	711	-8	4	Clothing	Saree	Clothing-Saree	2844
B-25635	382	30	3	Clothing	Saree	Clothing-Saree	1146
B-25636	637	113	5	Clothing	Saree	Clothing-Saree	3185
B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	488
B-25646	20	-8	2	Clothing	Saree	Clothing-Saree	40
B-25647	42	-6	4	Clothing	Saree	Clothing-Saree	168
B-25648	55	-26	4	Clothing	Saree	Clothing-Saree	220

Table: Order Details (1,500 rows) Column: Revenue (845 distinct values)

Data

- List of Orders
 - City
 - CustomerName
 - Order Date
 - Order ID
 - State
 - Order Details
 - Amount
 - Category
 - Category Type
 - Order ID
 - Profit
 - Quantity
 - Revenue
 - Sub-Category
 - Sales target

Revenue = 'Order Details'[Amount] * 'Order Details'[Quantity]

Calculated **Revenue** per Order in the Order Details Table, Created a calculated column in the Order Details table that calculates the revenue (**Amount × Quantity**) for each order.

- **Create a Calculated Column to Categorize Sales:** Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

File Home Help Table tools Column tools Share

Name **Order Details** Manage relationships New measure measure column New table New table Mark as date table

Structure Relationships Calculations Calendars

1 Sales Category = IF('Order Details'[Revenue]>CALCULATE(AVERAGE('Order Details'[Revenue]),ALL('Order Details')),"Above Average","Below Average")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing-Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing-Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing-Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing-Saree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing-Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing-Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing-Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing-Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing-Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing-Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing-Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing-Saree	1146	Below Average
B-25636	637	113	5	Clothing	Saree	Clothing-Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing-Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing-Saree	168	Below Average

Data

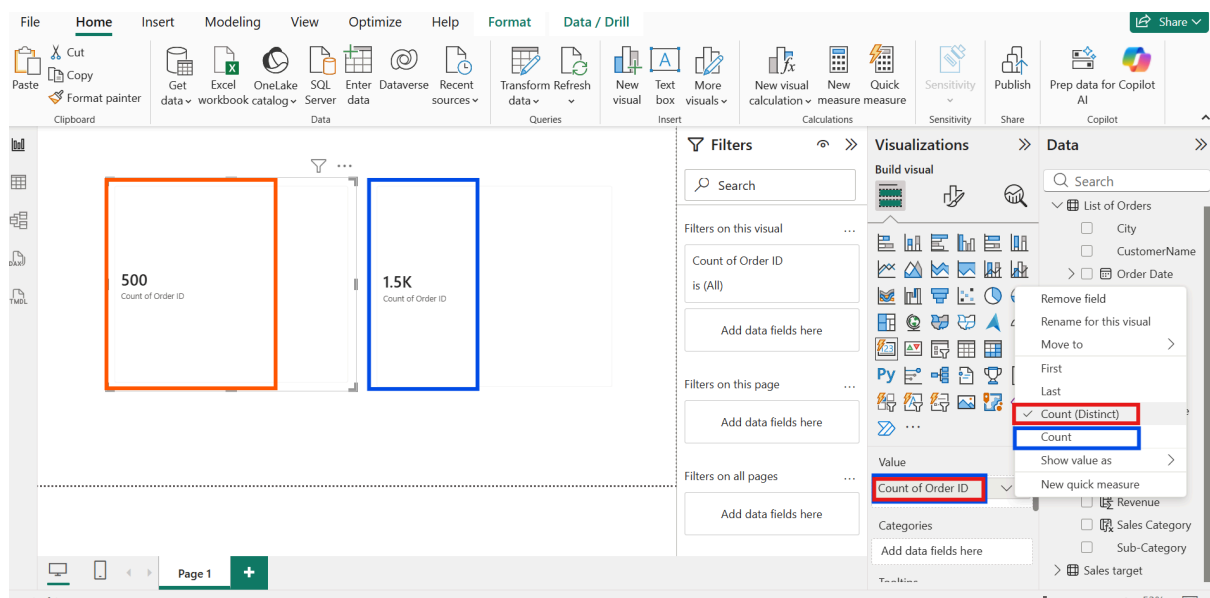
- List of Orders
 - City
 - CustomerName
 - Order Date
 - Order ID
 - State
 - Order Details
 - Amount
 - Category
 - Category Type
 - Order ID
 - Profit
 - Quantity
 - Revenue
 - Sales Category
 - Sub-Category
 - Sales target

Sales Category = IF('Order Details'[Revenue]>CALCULATE(AVERAGE('Order Details'[Revenue]),ALL('Order Details')),"Above Average","Below Average")

Sales Category is Created a calculated column using the **IF**, **CALCULATE**, **AVERAGE**, and **ALL** functions to compare each order's **Revenue** with the overall average amount. Orders with an amount greater than the average were categorized as "**Above Average**", while the remaining orders were categorized as "**Below Average**".

Calculated Measures

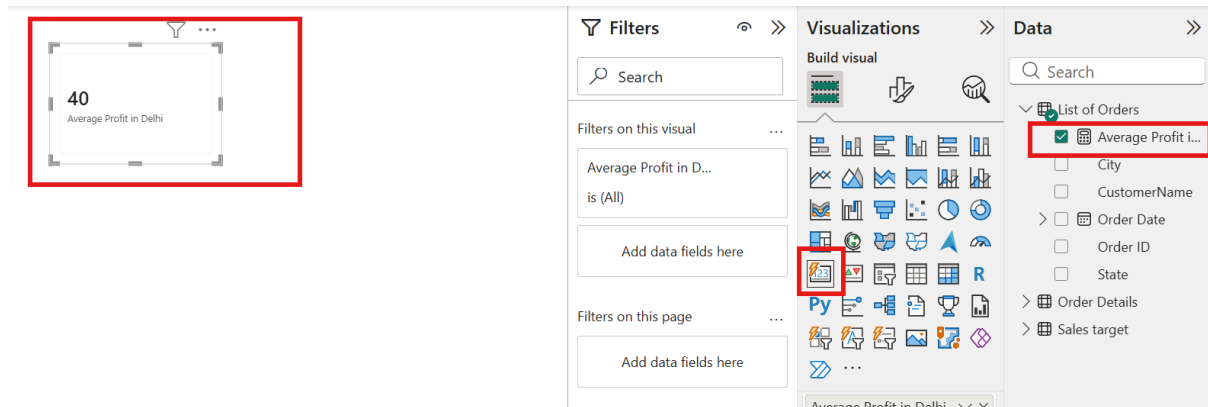
1. Order Count- Create a measure that calculates the total number of orders in the Order Details table.



Card 1: **Unique Order Count (DISTINCTCOUNT)** → Displays the total number of unique orders.

Card 2: **Order Count (COUNT)** → Displays the total number of order records (rows).

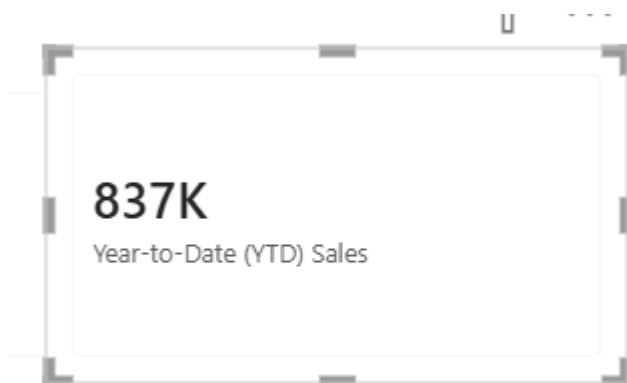
2. Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.



Average Profit in Delhi = `CALCULATE(AVERAGE('Order Details'[Profit]), 'List of Orders'[State] = "Delhi")`

Average Profit in Delhi: Created a measure to calculate the average profit for orders placed in Delhi and displayed the result in a **Card** visual.

3. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.

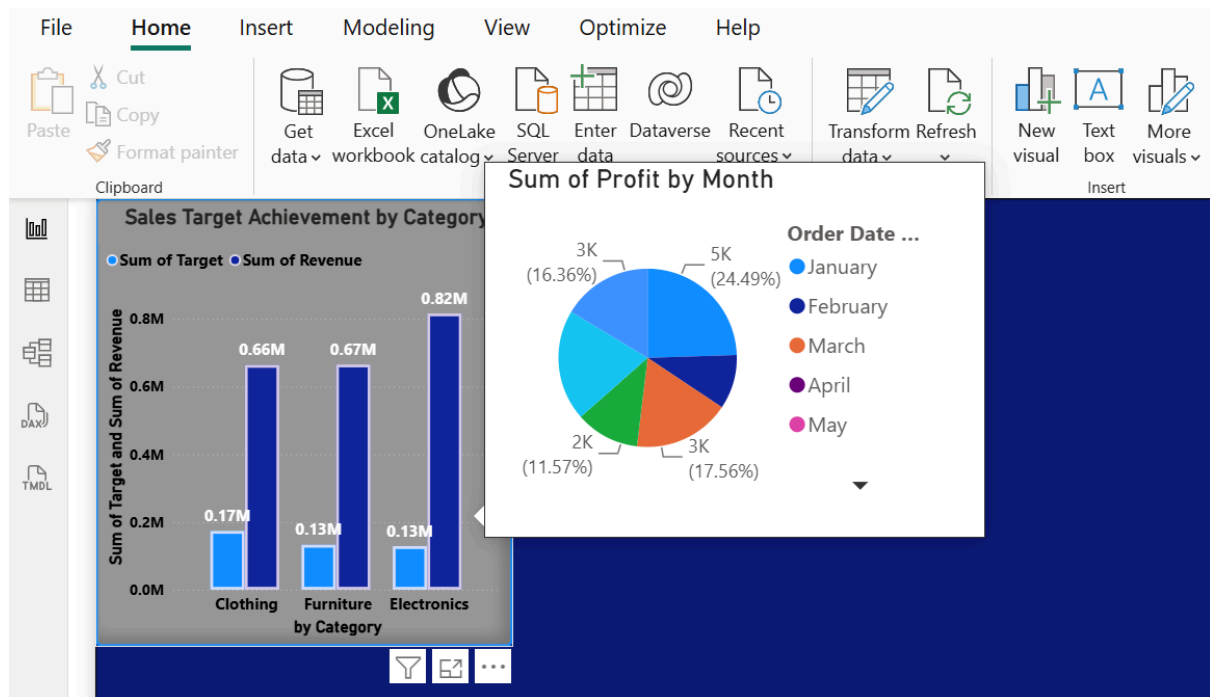


Year-to-Date (YTD) Sales = `CALCULATE(SUM('Order Details'[Revenue]), DATESYTD('List of Orders'[Order Date]))`

Year-to-Date (YTD) Sales: Created a measure using the **CALCULATE**, **SUM**, and **DATESYTD** functions to calculate the total sales accumulated from the beginning of the year up to the current order date. Displayed the result using a **Card** visual.

Create the following visualizations using Power BI:

- **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.



Sales Target Achievement by Category: Created a **Clustered Column Chart** to compare **Actual Sales** with **Sales Targets** across different categories. Added a **report page tooltip** displaying **Order Date** and **Profit** to provide additional insights when hovering over the chart.

- **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.

File Home Help Table tools Column tools

Name: Profit Margin Data type: Whole number Format: General Summarization: Sum Data category: Uncategorized

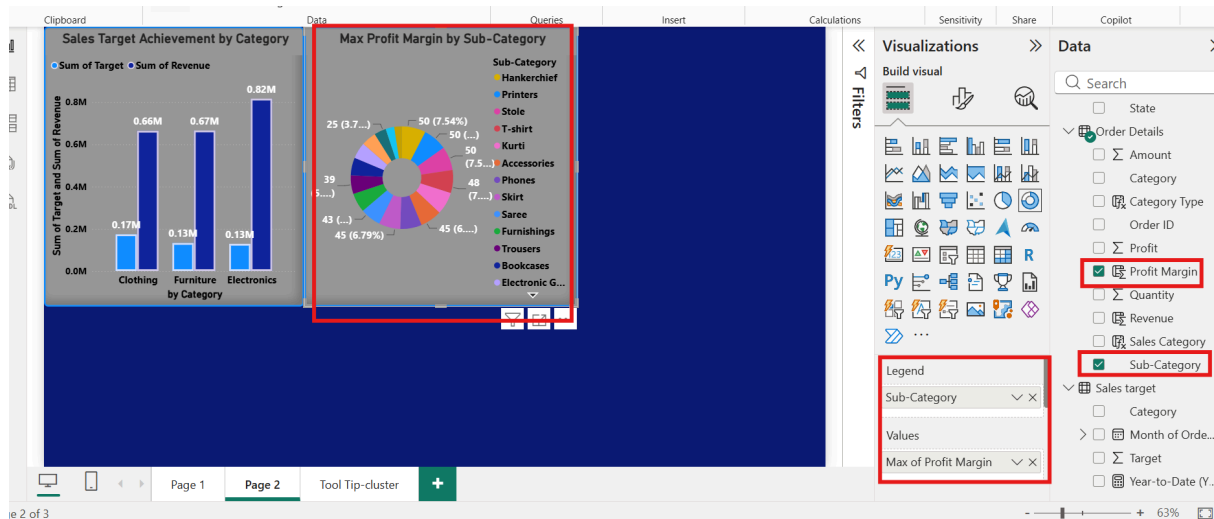
1 Profit Margin = DIVIDE('Order Details'[Profit], 'Order Details'[Revenue], 0) * 100

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category	Profit Margin
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683	Above Average	13
B-25602	119	-5	8	Clothing	Saree	Clothing-Saree	952	Below Average	-1
B-25603	193	-166	3	Clothing	Saree	Clothing-Saree	579	Below Average	-29
B-25604	157	5	9	Clothing	Saree	Clothing-Saree	1413	Below Average	0
B-25605	75	0	7	Clothing	Saree	Clothing-Saree	525	Below Average	0
B-25609	25	-5	4	Clothing	Saree	Clothing-Saree	100	Below Average	-5
B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129	Below Average	0
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320	Below Average	-18
B-25613	1603	0	9	Clothing	Saree	Clothing-Saree	14427	Above Average	0
B-25619	353	90	8	Clothing	Saree	Clothing-Saree	2824	Above Average	3
B-25622	534	0	3	Clothing	Saree	Clothing-Saree	1602	Above Average	0
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596	Below Average	-15
B-25625	635	-349	5	Clothing	Saree	Clothing-Saree	3175	Above Average	-11
B-25628	24	-9	4	Clothing	Saree	Clothing-Saree	96	Below Average	-9
B-25633	711	-8	4	Clothing	Saree	Clothing-Saree	2844	Above Average	0
B-25635	382	30	3	Clothing	Saree	Clothing-Saree	1146	Below Average	3
B-25636	637	113	5	Clothing	Saree	Clothing-Saree	3185	Above Average	4
B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	488	Below Average	-10
B-25646	20	-8	2	Clothing	Saree	Clothing-Saree	40	Below Average	-20
B-25647	42	-6	4	Clothing	Saree	Clothing-Saree	168	Below Average	-4
B-25648	55	-26	4	Clothing	Saree	Clothing-Saree	220	Below Average	-12

Table: Order Details (1,500 rows) Column: Profit Margin (109 distinct values)

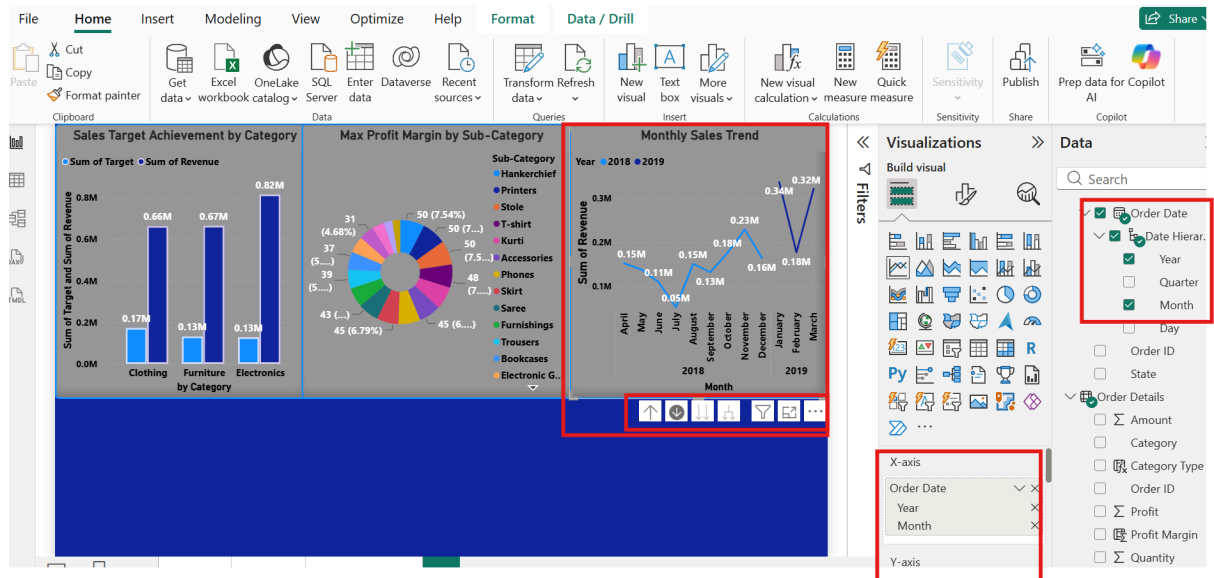
Data pane: List of Orders, Order Details, Amount, Category, Category Type, Order ID, Profit, Revenue, Sales Category, Sub-Category, Sales target, Category, Month of Order Date, Target, Year-to-Date (YTD) Sales

Profit Margin = DIVIDE('Order Details'[Profit], 'Order Details'[Revenue], 0) * 100



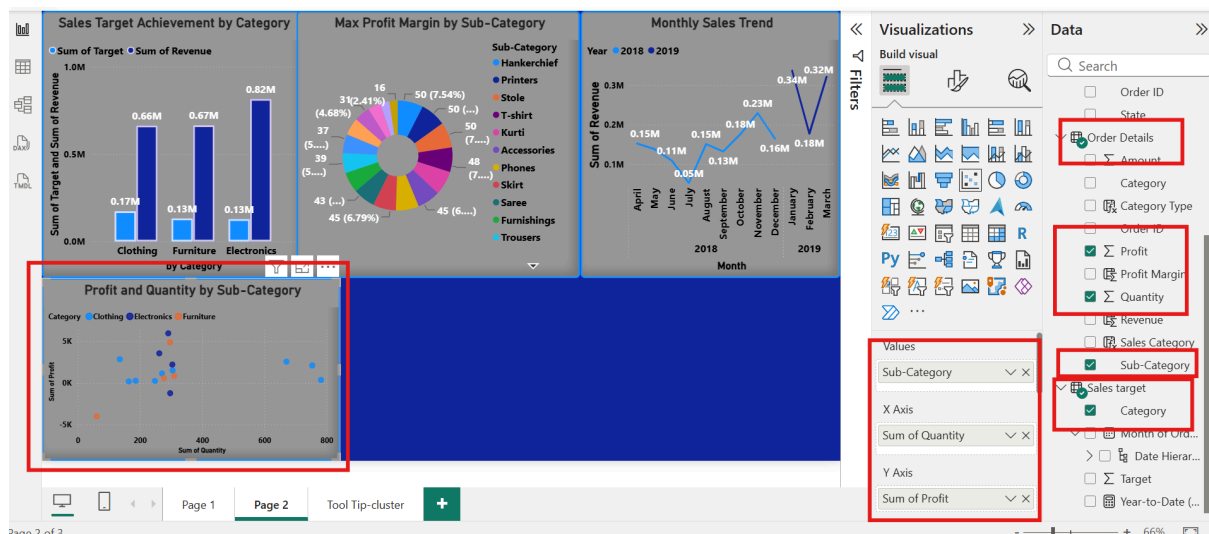
Max Profit Margin by Sub-Category: Created a **Profit Margin** calculated column using the **Profit** and **Revenue** fields. Developed a **Donut Chart** to display the **maximum profit margin** for each product **sub-category** by applying Maximum summarization to the Profit Margin field.

- **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.



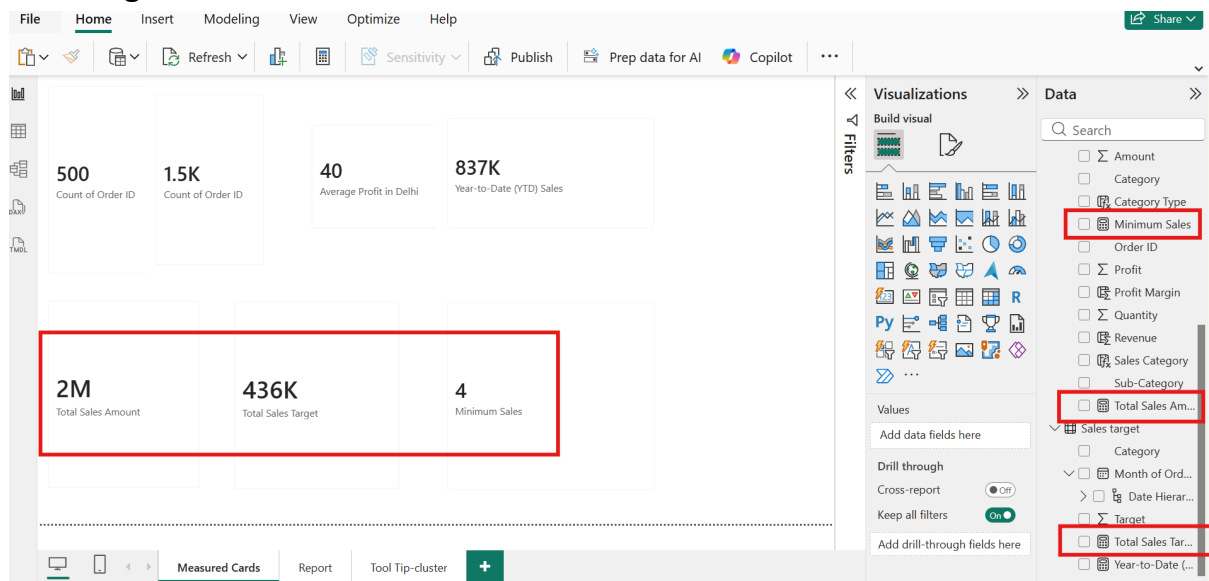
Used a Line Chart to show the monthly sales trend. Order Date was added to the X-axis, Sum of Revenue to the Y-axis, and Year to the Legend. Used the hierarchy drill-down option to view sales month-wise

- **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

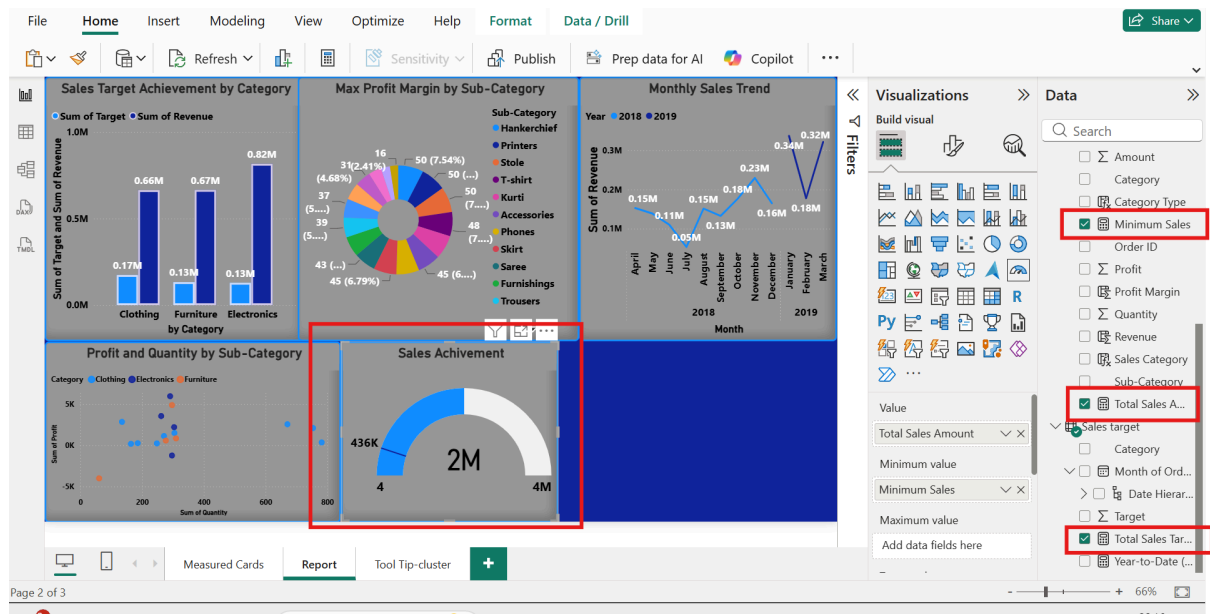


Used a Scatter Chart to compare Profit and Quantity by Sub-Category. Added **Sum of Quantity** to the X-axis, **Sum of Profit** to the Y-axis, **Sub-Category** to the **Values**, and **Category** to the **Legend**. The chart shows each category in different colors.

- **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

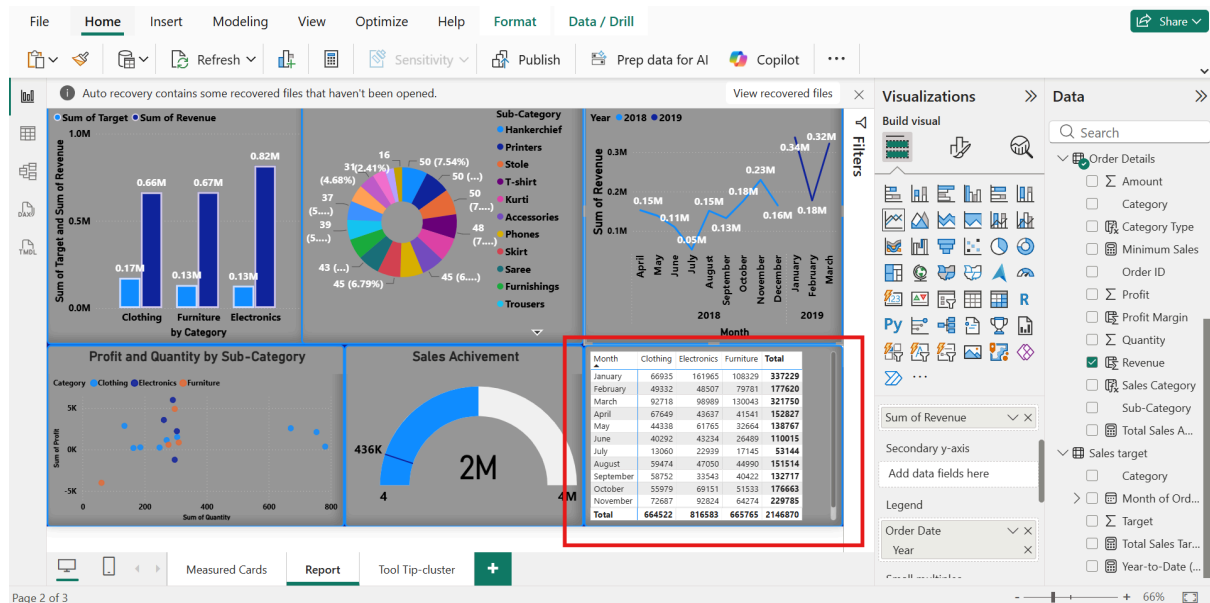


Created two cards: one for the **Sum of Revenue** and another for the **Sum of Sales Target** and I have created one more card for **Minimum of sales** From Revenue



Created a **Gauge Visual** to compare Total Sales with the Sales Target. That's why I have created one more measures card for minimum sales.

- **Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.**

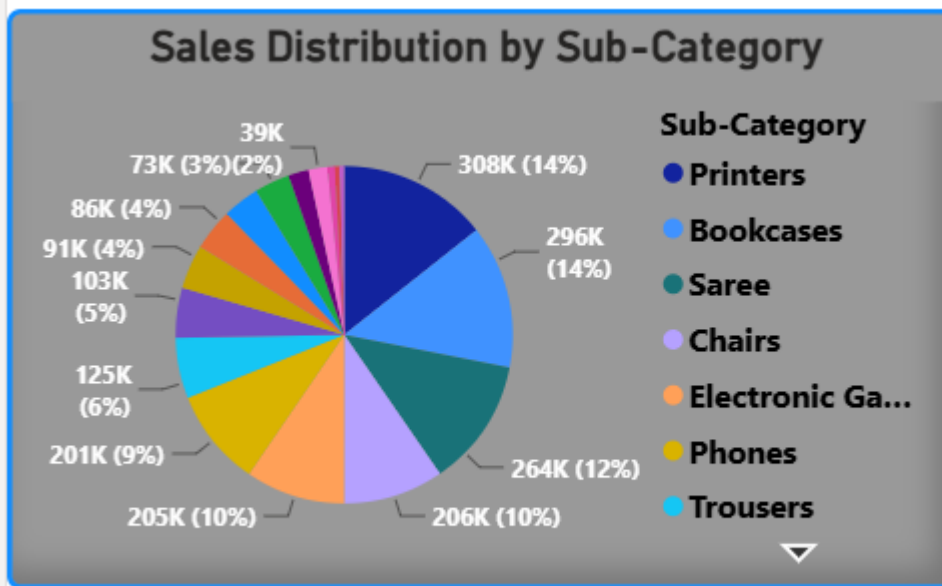


Created a Matrix visual to display monthly sales by category. Added **Month** to the Rows, Created a Matrix visual to display monthly sales by category. Added Month to the Rows, Category to the Columns, and Sum of Revenue to the Values. to the Columns, and **Sum of Revenue** to the Values.

- **Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.**

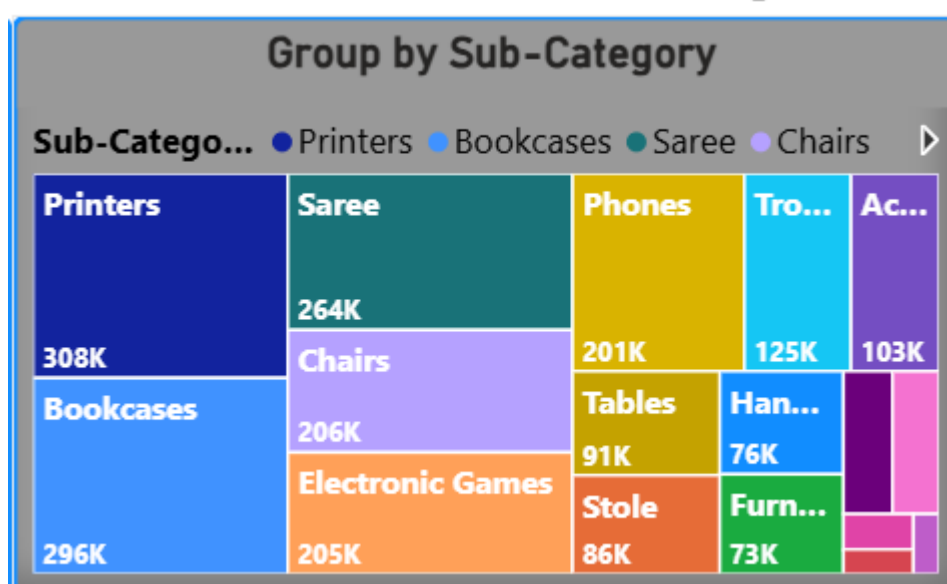
I cant able to create Map, my lap global settings are disabled

- **Sales Distribution by Sub-Category:** Represent the sales distribution across different



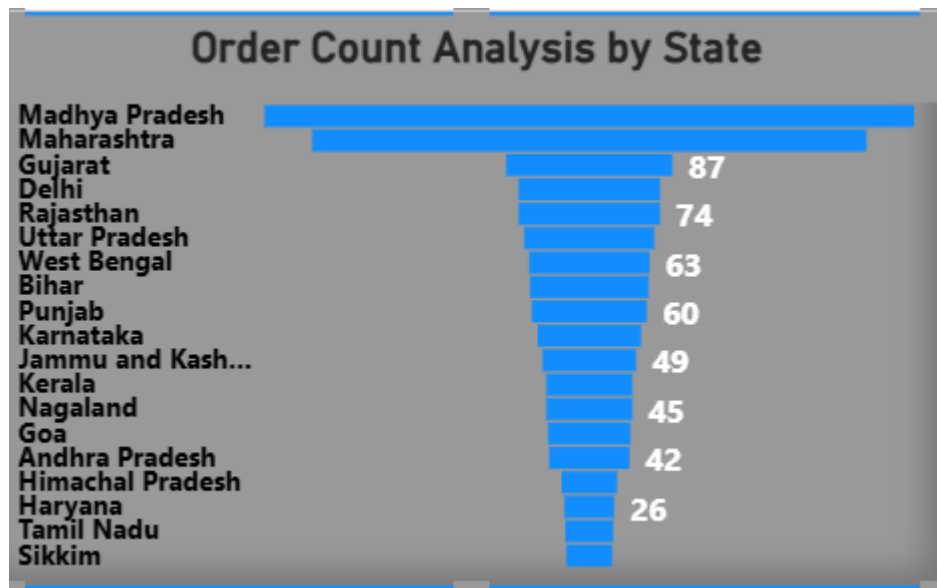
Created a Pie Chart to show the sales distribution by Sub-Category. Added **Sub-Category** to the **Legend** and **Sum of Revenue** to the **Values**.

- **sub-categories using a treemap.**



Created a Treemap to show the sales distribution by Sub-Category. Added **Sub-Category** to the **Group** and **Sum of Revenue** to the **Values**

- **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.



Created a Funnel Chart to show the order count by state. Added State to the Category and Count of Order ID to the Values.